

Network-Based Simulation of Flood Alert Dissemination Using GDACS and Synthetic Communication Logs

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Abstract

This study presents a simulation-based analysis of disaster alert dissemination using flood alerts from the Global Disaster Alert and Coordination System (GDACS) combined with a synthetic stakeholder communication network. The proposed framework models information flow across control centers, media outlets, community leaders, and citizens using the Independent Cascade Model. A directed graph was constructed from 300 simulated message exchanges, incorporating delay and reliability attributes. Results from ten simulation runs show significant variability in alert reach (19–49%) and propagation delays, with citizens experiencing the longest average delay (348 sec). Centrality and bottleneck-based analysis identified mid-network actors, particularly community leaders and media outlets, as critical to effective dissemination. These findings underscore the structural dependencies within early warning systems and highlight the need for resilient communication pathways in disaster response networks.

CCS Concepts

• **Applied computing**; • **Computing methodologies**; • **Social and professional topics**;

Keywords

Ubiquitous Computing, Early Warning Systems, Information Diffusion, Stakeholder Networks, Alert Propagation, Emergency Response Modeling

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1 Introduction

Timely and reliable dissemination of emergency early warnings is essential for minimizing human and material losses during disasters. The Global Disaster Alert and Coordination System (GDACS) serves as a central platform for issuing real-time alerts on global natural hazards, including floods [13]. These alerts are distributed through SMS, email, and increasingly via social media and mobile apps to enhance reach and timeliness [13]. Notably, early studies by the Joint Research Centre found that platforms like Twitter could provide critical situational information, such as infrastructure damage, within the first 30 minutes of a disaster, often preceding conventional media reports [13]. Despite GDACS's strengths in detection and coordination, the mechanisms by which alerts traverse stakeholder hierarchies from institutional entities to at risk communities remain insufficiently understood. To address this, we propose a simulation-based framework to model flood alert dissemination across a multi-tiered network, with a focus on recent European events. Previous research highlights limitations in current mobile emergency applications, including low user engagement and fragmented interoperability across agencies, particularly in Nordic countries where cross-border cooperation is vital [12]. This underscores the need for multi-stakeholder dissemination models that integrate both infrastructure dependencies and user behavior.

Recent advancements in ubiquitous computing have introduced alternative mechanisms to enhance alerting systems. Robinson et al. [10] advocated for inclusive communication protocols by enabling assistance dogs to initiate emergency alerts, while Hasan et al. [4] demonstrated the use of Bluetooth beacons for rapid intra-city alert propagation. In low-connectivity environments, Lee et al. [8] employed social-context-aware routing via the *ChitChat* system, and studies by Tran et al. [15] and Huang [5] proposed UAV-based and hybrid network solutions to enhance resilience and energy efficiency in crisis communications. These studies inform the architecture of resilient, inclusive dissemination frameworks. The

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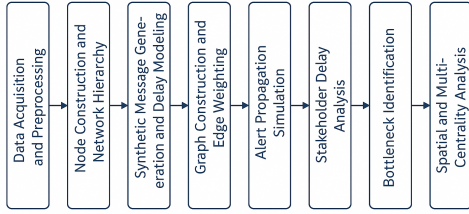


Figure 1: Methodological workflow for simulating and analyzing flood alert propagation using GDACS data and synthetic stakeholder networks.

objective of this research is to simulate a synthetic message-passing network that emulates real-world alert dissemination pathways. Stakeholders—including control centers, media outlets, community leaders, and citizens—are represented as nodes, with directed edges encoding transmission delays and reliability. We adopt diffusion models such as the Independent Cascade (IC) and threshold activation to assess network efficiency, equity, and robustness [14]. The use of synthetic data is methodologically warranted, given the challenges of accessing operational alert logs due to privacy and political constraints [1]. Synthetic datasets allow for realistic yet controllable simulations that include communication latency, stakeholder hierarchies, and reliability metrics [11].

We hypothesize that network topology significantly affects alert timeliness and completeness. While highly connected hubs like media outlets can expedite propagation, structurally critical nodes with high betweenness and delay can hinder it. Simulation outputs and network metrics are used to identify such bottlenecks. Moreover, we posit that channels with higher reliability, such as mobile apps with geolocation, offer superior performance over legacy systems like SMS. This framework enables actionable insights for emergency planners, allowing redesign of communication chains, mitigation of weak points, and spatially targeted interventions. Ultimately, this study offers a reproducible method to evaluate and optimize early warning dissemination, contributing to broader goals in disaster risk reduction and resilience.

2 Related Work

Traditional studies have explored various communication channels and their effectiveness in different emergency scenarios. Zang et al. [17] conducted simulations comparing six distinct information dissemination media, such as SMS, microblogs, news portals, cell phones, television, and oral communication, highlighting that no single medium is universally optimal. Instead, a hybrid model tailored to the emergency type and community context yields the most effective communication outcomes. Perera et al. [9], in their comprehensive review of global flood early warning systems (FEWS), identified technical and institutional barriers to effective communication, such as insufficient stakeholder coordination and limited community engagement. Their findings underscore the importance of designing inclusive, community-centered systems that integrate

local knowledge and promote active participation. The integration of artificial intelligence (AI) and machine learning (ML) into early warning systems has been recognized for enhancing predictive capabilities [12]. The IEEE Public Safety Technology Initiative [6] emphasizes the role of predictive analytics and ML in improving early warning systems, outlining key components such as data collection from diverse sources and model development using algorithms like neural networks and support vector machines.

Recent advancements in ubiquitous computing have explored innovative methods for emergency alert dissemination. For instance, Kumar et al. [7] propose enhancements to Wireless Emergency Alerts, aiming to improve their effectiveness in crises. Similarly, Yu et al. [16] develop a privacy-preserving indoor monitoring system that detects human presence through ground vibrations, ensuring timely alerts even when traditional infrastructures fail. Additionally, Guerrero et al. [3] designed an IoT-based warning system for rural areas, facilitating rapid evacuations during natural disasters. These studies underscore the critical role of ubiquitous technologies in enhancing emergency communication networks. However, the development and testing of such advanced systems often face challenges due to the scarcity of high-quality, real-world data. Synthetic data emerges as a viable solution to this problem. It allows for the simulation of various scenarios, enabling researchers to test and validate models under controlled conditions without the ethical and logistical constraints associated with real data collection. Synthetic data can mimic the statistical properties of real-world data, providing a safe and efficient means to train and evaluate early warning systems [2]. In the context of emergency communication networks, synthetic data facilitates the modeling of complex stakeholder interactions and information flows. By simulating message propagation through various channels and nodes, researchers can identify potential bottlenecks and optimize communication strategies. This approach aligns with the objectives of the current study, which aims to model the information diffusion process, assess delays, and identify central actors in the flow of warnings using synthetic datasets derived from GDACS flood alerts.

3 Simulation of Flood Alert Dissemination

This study presents a simulation-based framework to model and analyze the propagation of disaster alerts through a stakeholder communication network, using real-world flood alert data from the Global Disaster Alert and Coordination System (GDACS) and a synthetically generated diffusion log. The primary objective is to capture structural dependencies, propagation delays, and message reliability across hierarchical actors in the disaster alert ecosystem.

3.1 Data Preprocessing and Network Construction

Flood event data for the past two years was retrieved via the *gdacs-api* Python library. The dataset was filtered to retain only flood events located in European countries. Country-level filtering was performed by matching the GDACS country field against a pre-compiled list of European nations, while temporal filtering utilized the event publication date. The alert origin nodes were assigned the actual latitude and longitude values retrieved from GDACS to

preserve spatial authenticity in the simulation model. The simulation network consisted of five distinct stakeholder types, such as alert origin, control centers, media outlets, community leaders, and citizens. Each node type was defined by four parameters: count, role priority, communication capacity, and influence score. These parameters simulate real-world communication capabilities and behavioral tendencies. For example, alert origin nodes were configured with a maximum influence score of 1.0 and high capacity, whereas citizens had a lower influence score of 0.2 and limited capacity. Latitude and longitude coordinates were randomly assigned within European bounds, except for alert origin nodes which inherited coordinates directly from GDACS. These configurations enabled role-aware propagation and control over simulation dynamics.

3.2 Synthetic Message Generation and Delay Modeling

To emulate realistic communication flows, a synthetic diffusion dataset comprising 300 message records was generated. Message propagation followed a predefined four-stage hierarchical path: from GDACS to control centers, then to media outlets, community leaders, and finally citizens. Each message included attributes such as message type (SMS or App Notification), delay time (randomized between 10 and 180 seconds), and reliability score (random float between 0.00 and 1.00). A brief processing delay (10–60 seconds) was added between node transmissions to increase realism. A directed graph was constructed using the communication log and node metadata. Each edge represented a message transmission and carried three primary attributes, namely average delay, average reliability, and message count. Edge weights were calculated using the following Equation 1,

$$\text{weight} = \frac{\text{average delay}}{\text{average reliability} + 1 \cdot 10^{-6}} \quad (1)$$

This formulation penalizes edges with low reliability and high delays, simulating poor communication links. The additive constant prevents division by zero. This function effectively emphasizes weak links, thereby influencing centrality scores and activation likelihood in the propagation model. Normalized values of delay and reliability were also computed using Equation 2. This min-max normalization scaled both attributes to the $[0,1]$ interval, facilitating their use in network visualization and comparative analysis.

$$X' = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (2)$$

The simulation of 300 synthetic messages along a hierarchical alert path yielded average delay and reliability values (94.5s and 0.52, respectively), reflecting a balanced dissemination pattern. SMS slightly outnumbered App notifications, aligning with real-world practices. Communication from community leaders to citizens showed the lowest delays, while control center to media outlet links were the slowest, suggesting potential congestion. Reliability was highest from the alert origin to control centers (0.59) and lowest from media outlets to community leaders (0.46), indicating midstream weaknesses. The resulting network of 181 nodes and 262 edges formed a weakly connected, directed graph suitable for cascade modeling. Multiple connections, especially from the

alert origin to control centers, reflect realistic redundancy. Edge weights computed as delay-to-reliability ratios exhibited significant variance, with extreme values from media-to-community links driven by near-zero reliability. These results highlight structural bottlenecks and the critical role of reliability in ensuring timely and trustworthy alert propagation. On average, communication from alert origins to control centers exhibited the lowest edge weights, reinforcing their role as high-reliability, low-latency communication anchors. Conversely, the high edge weights in the media-to-community segment suggest a potential structural bottleneck, exacerbated by lower reliability and higher delays. These findings justify the use of delay-normalized reliability scoring in simulation-based modeling to capture nuanced trade-offs between speed and trustworthiness in alert dissemination.

3.3 Alert Propagation and Stakeholder Delay Simulation

The Independent Cascade (IC) model was employed to simulate the alert spread. Starting from the GDACS node at time $t = 0$, each active node had a single opportunity to activate its neighbors with a probability equal to the edge reliability. Delay attributes determined the time of each activation event. A queue-based event processing algorithm was used to schedule and log activation attempts, with propagation terminating when the queue was exhausted. Activation times were recorded per node, forming the basis for delay analysis. The alert reception times were grouped by stakeholder type. Delay metrics (mean, min, max) were computed using pandas group-by operations. This analysis highlighted how quickly different stakeholder categories received alerts and revealed systemic disparities, such as media outlets receiving alerts faster than community-level actors. The resulting bar plots enabled visual inspection of communication efficiency by role. Communication bottlenecks were identified through dual-criteria filtering. First, nodes with propagation delays above the 60th percentile were labeled as high-delay nodes. Second, betweenness centrality (weighted by delay) was computed using Equation 3. Nodes exceeding the 60th percentile in centrality were marked as high centrality. Intersections of these sets were identified as bottlenecks nodes that were both structurally vital and slow in receiving or forwarding alerts. These were visualized using node-size and color-coded network plots.

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (3)$$

An interactive Folium map was generated to provide a spatial view of alert propagation. Nodes were colored by type and scaled by normalized betweenness centrality, calculated as shown in Equation 4,

$$r = 5 + X' \cdot 20 \quad (4)$$

Whereas X' is the normalized centrality. This allowed spatial inspection of influence and propagation bottlenecks. To identify globally critical nodes, the following centrality measures were calculated: degree, closeness (delay-weighted), betweenness (delay-weighted), and eigenvector centrality (using edge weights). Nodes ranking in the top 15 across at least three measures were labeled

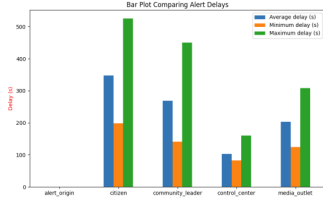


Figure 2: Bar plot comparing alert delays between stakeholder types.

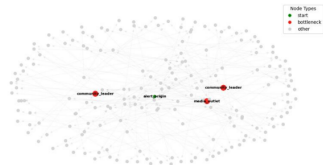


Figure 3: Network graph highlighting potential bottlenecks with 60th percentile thresholds.

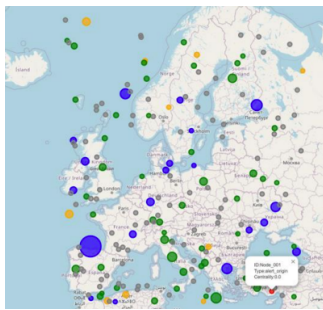


Figure 4: Map highlighting the alert origin and other subsequent alert stations.

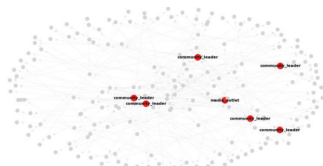


Figure 5: Critical nodes based on modified centrality rankings.

as structurally critical. This multi-metric assessment provided robustness against single-criterion bias and highlighted systemic vulnerabilities.

Simulation results using the Independent Cascade model exhibited considerable variability in alert dissemination, with alert reach ranging from 19.34% to 48.62% and an average of 35.31%, refer to Table 1. This variability reflects the probabilistic nature of the model and its sensitivity to edge reliability values. Propagation delays followed a hierarchical structure, with control centers receiving alerts fastest (average delay: 102.4 s) and citizens the slowest (average delay: 348 s), as shown in the bar plot in Figure

2. This pattern aligns with the network’s layered design, where stakeholders closer to the alert origin naturally experience lower latencies. The spatial distribution of central actors is visualized in Figure 4, where node size reflects betweenness centrality and color indicates stakeholder type. Media outlets and community leaders appear prominently, positioned near the center of the network, confirming their strategic roles in message propagation. Figure 3 highlights three critical nodes, two community leaders, and one media outlet that simultaneously exhibit high propagation delays and high betweenness centrality under 60th percentile thresholds. These nodes represent structural bottlenecks whose performance heavily influences downstream alert delivery. Furthermore, the multi-centrality analysis shown in Figure 5 identifies a cluster of nodes, predominantly community leaders, that consistently rank in the top 15 for degree, closeness, and betweenness centrality. Their placement across multiple metrics reinforces their role as pivotal connectors within the network. Together, these figures illustrate that the alert system’s effectiveness is contingent upon a small subset of mid-network actors. Their delay or failure could severely compromise timely and wide-scale alert dissemination.

Table 1: Summary statistics of 10 simulation runs.

Sim. No.	Nodes Activated	Reach	Avg. Time (s)	Events
1	75	41.44%	263.16	188
2	52	28.73%	260.11	147
3	65	35.91%	275.06	193
4	84	46.41%	307.97	213
5	46	25.41%	231.27	99
6	35	19.34%	267.12	85
7	88	48.62%	284.91	225
8	80	44.20%	270.28	196
9	60	34.81%	293.68	158
10	51	28.18%	276.05	150
Average	63.90	35.31%	272.96	162.40

4 Conclusion

This study modeled GDACS-based flood alert propagation through a synthetic multi-stakeholder network, revealing the critical role of mid-network actors in ensuring timely and broad alert dissemination. Simulation results underscore structural vulnerabilities, particularly at community leader and media outlet nodes. Future work will explore real-time behavioral data, alternative diffusion models, and integration with geospatial risk indicators to enhance the framework’s realism and applicability for operational early warning systems.

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References

- [1] Dmytro Chumachenko. 2025. Machine Learning Approaches for Infectious Disease Surveillance During Emergencies. In *Developments in Information and Knowledge Management Systems for Business Applications: Volume 8*. Springer, 555–591.

- [2] CIO Editors. 2025. Synthetic Data's Fine Line Between Reward and Disaster. <https://www.cio.com/article/3986687/synthetic-datas-fine-line-between-reward-and-disaster.html>. Accessed: 2025-05-30.
- [3] Diego Ricardo Cajachagua Guerreros, Shirley Gabriela Cardenas Quispe, Paul Cristian Guerreros Abregu, Mohamed Mehdi Hadi Mohamed, and Sario Angel Chamorro Quijano. 2022. Design of an Automatic Landslide and Earthquake Warning System in Rural Areas of Peru. In *2022 IEEE 13th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*. IEEE, 0303–0307.
- [4] Raiful Hasan, Ragib Hasan, and Tanveer Islam. 2021. InSight: A Bluetooth beacon-based ad-hoc emergency alert system for smart cities. In *2021 IEEE 18th Annual Consumer Communications & Networking Conference (CCNC)*. IEEE, 1–6.
- [5] Zhengrui Huang. 2023. Hybrid device-to-device and device-to-vehicle networks for energy-efficient emergency communications. *IEEE Systems Journal* 17, 4 (2023), 5429–5440.
- [6] IEEE Public Safety Technology Initiative. 2024. Predictive Analytics in Disaster Prevention: Machine Learning Models for Early Warning Systems. <https://publicsafety.ieee.org/topics/predictive-analytics-in-disaster-prevention-protect/penalty/z@machine-learning-models-protect/penalty/z@for-early-warning-systems/>. Accessed: 2025-05-30.
- [7] Sumet Kumar, Hakan Erdogmus, Bob Iannucci, Martin Griss, and João Diogo Falcão. 2018. Rethinking the future of wireless emergency alerts: A comprehensive study of technical and conceptual improvements. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 2 (2018), 1–33.
- [8] Douglas McGeehan, Sanjay Madria, and Dan Lin. 2020. Effective social-context based message delivery using ChitChat in sparse delay tolerant networks. *Distributed and Parallel Databases* 38, 2 (2020), 401–438.
- [9] Duminda Perera, Jetal Agnihotri, Ousmane Seidou, and Riyanti Djalante. 2020. Identifying societal challenges in flood early warning systems. *International Journal of Disaster Risk Reduction* 51 (2020), 101794.
- [10] Charlotte L Robinson, Clara Mancini, Janet Van Der Linden, Claire Guest, and Robert Harris. 2014. Canine-centered interface design: supporting the work of diabetes alert dogs. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 3757–3766.
- [11] Wasswa Shafik. 2024. An Overview of Computational Modeling and Simulations in Wireless Communication Systems. *Computational Modeling and Simulation of Advanced Wireless Communication Systems* (2024), 8–40.
- [12] Fuzel Ahamed Shaik and Mourad Oussalah. 2024. On mining mobile emergency communication applications in Nordic countries. *International Journal of Disaster Risk Reduction* 108 (2024), 104566.
- [13] Beate Stollberg and Tom De Groeve. 2012. The use of social media within the global disaster alert and coordination system (GDACS). In *Proceedings of the 21st International Conference on World Wide Web*. 703–706.
- [14] Leonidas Theodorakopoulos, Alexandra Theodoropoulou, and Christos Klavdiannos. 2025. Interactive Viral Marketing Through Big Data Analytics, Influencer Networks, AI Integration, and Ethical Dimensions. *Journal of Theoretical and Applied Electronic Commerce Research* 20, 2 (2025), 115.
- [15] Dinh-Hieu Tran, Van-Dinh Nguyen, Symeon Chatzinotas, Thang X Vu, and Björn Ottersten. 2021. UAV relay-assisted emergency communications in IoT networks: Resource allocation and trajectory optimization. *IEEE Transactions on Wireless Communications* 21, 3 (2021), 1621–1637.
- [16] Yang Yu and Torben Weis. 2020. A privacy-protecting indoor emergency monitoring system based on floor vibration. In *Adjunct Proceedings of the 2020 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2020 ACM International Symposium on Wearable Computers*. 164–167.
- [17] Nan Zhang, Hong Huang, Boni Su, Jinlong Zhao, and Bo Zhang. 2014. Information dissemination analysis of different media towards the application for disaster pre-warning. *PloS one* 9, 5 (2014), e98649.